

Affine Caesar Cipher

5. Write a C program for generalization of the Caesar cipher, known as the affine Caesar cipher, has the following form: For each plaintext letter p , substitute the ciphertext letter C : $C = E([a, b], p) = (ap + b) \bmod 26$ A basic requirement of any encryption algorithm is that it be one-to-one. That is, if $p \neq q$, then $E(k, p) \neq E(k, q)$. Otherwise, decryption is impossible, because more than one plaintext character maps into the same ciphertext character. The affine Caesar cipher is not one-to-one for all values of a . For example, for $a = 2$ and $b = 3$, then $E([a, b], 0) = E([a, b], 13) = 3$.

a. Are there any limitations on the value of b ?

b. Determine which values of a are not allowed.

Aim

To write a C program to implement the **Affine Caesar Cipher** using the formula:

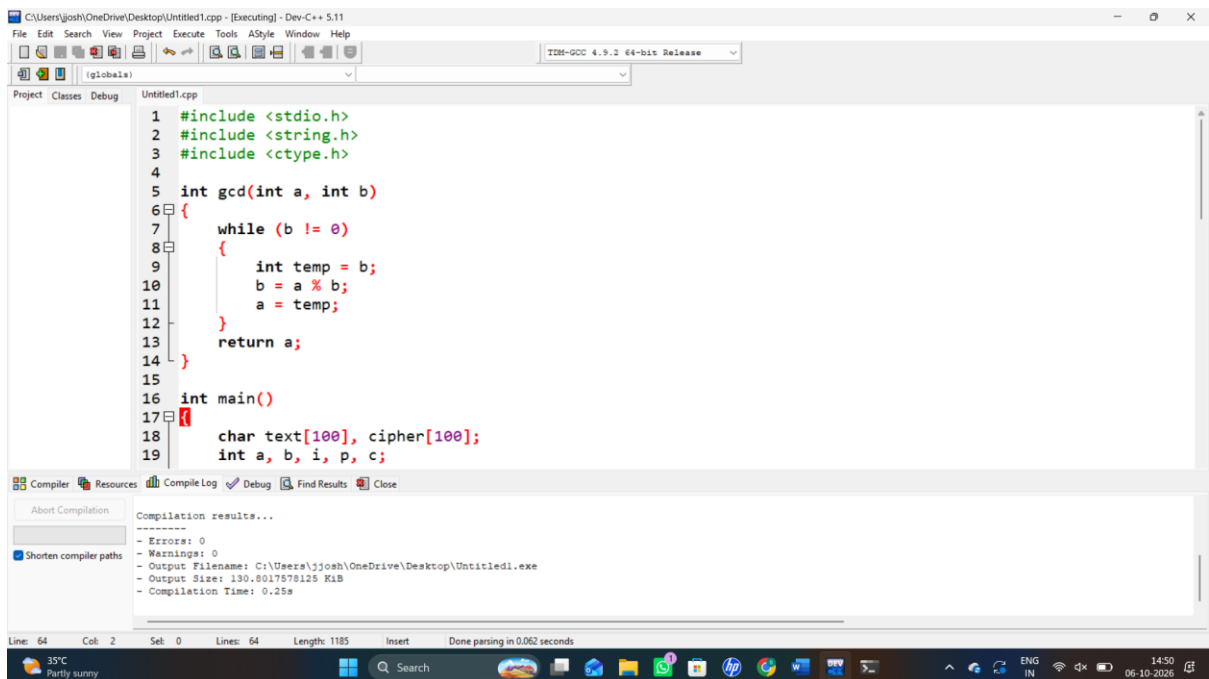
$$[C = (ap+b) \bmod 26]$$

and determine the restrictions on a and b .

Algorithm

1. Start the program.
2. Read the plaintext.
3. Read the values of a and b .
4. Check whether $\gcd(a, 26) = 1$.
5. If the GCD is not 1, display that a is invalid.
6. For every plaintext letter, convert it to a number from 0 to 25.
7. Apply: $[C = (ap+b) \bmod 26]$
8. Convert the result into a letter.
9. Display the ciphertext.
10. Stop.

Program

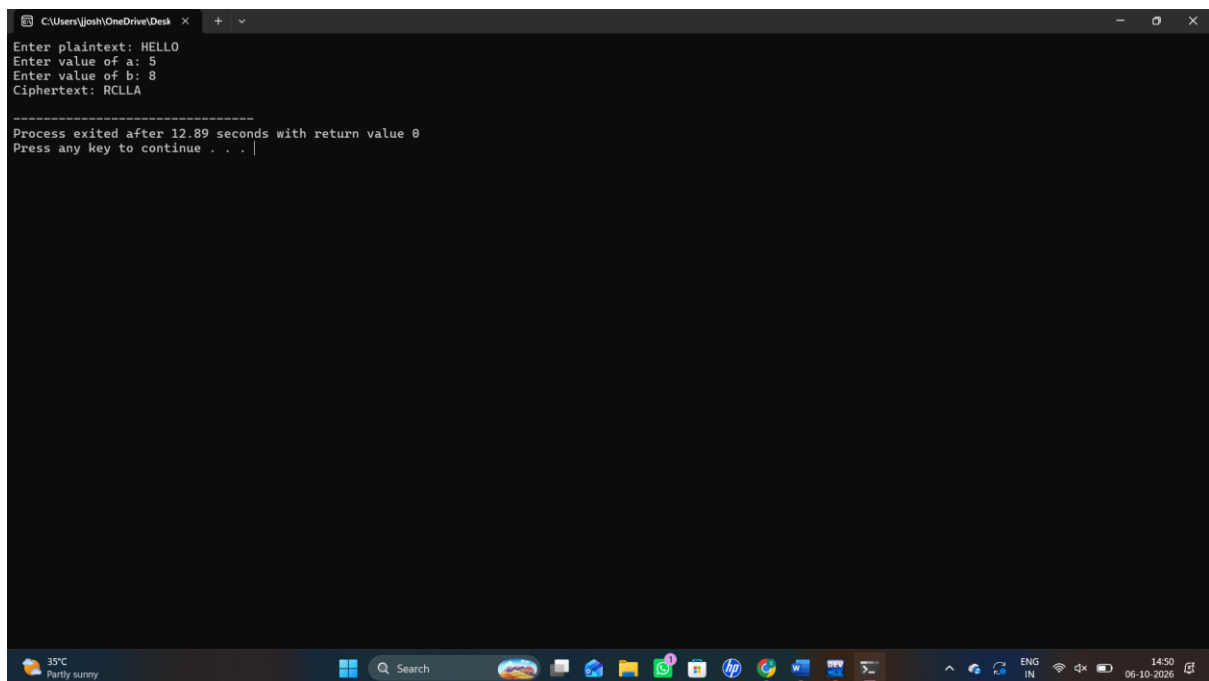


```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int gcd(int a, int b)
6 {
7     while (b != 0)
8     {
9         int temp = b;
10        b = a % b;
11        a = temp;
12    }
13    return a;
14 }
15
16 int main()
17 {
18     char text[100], cipher[100];
19     int a, b, i, p, c;
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled1.exe
- Output Size: 130.8017578125 KiB
- Compilation Time: 0.25s

Input/Output:



```
Enter plaintext: HELLO
Enter value of a: 5
Enter value of b: 8
Ciphertext: RCLLA

-----
Process exited after 12.89 seconds with return value 0
Press any key to continue . . .
```

a. Limitations on b:

There are no restrictions on b. Any value of b from 0 to 25 can be used.

b. Values of a not allowed:

For the cipher to be one-to-one:

$$[\gcd(a, 26) = 1]$$

Allowed values of a from 0 to 25 are:

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

Therefore, the not allowed values are:

0, 2, 4, 6, 8, 10, 12, 13, 14, 16, 18, 20, 22, 24

Result

Thus, the Affine Caesar Cipher was successfully implemented. The value of b has no restriction, while a must be relatively prime to 26.